Janeway’s OBGYN Association

Team Members:

Shani Plunkett-de la Cruz and Lon Danna

Project URL: http:// http://classwork.engr.oregonstate.edu:9116/

GitHub Repo Link: https://github.com/Lon20wn/cs340-database-project-team16

Executive Summary

Over the course of the 10-week term, our group developed and implemented an online database management system to be used by the fictional Janeway's OBGYN Association. This project included four major draft phases where class peers and teaching assistants reviewed our proposal, code, and UI then provided feedback on areas for improvement.

For the first step, we submitted our proposal for our database which included the overview, database outline, and the entity-relationship diagram. We proposed a database with five entities (Patients, AppointmentTypes, Appointments, Providers, and Clinics), all with varying attributes. Overall, our feedback was very positive with only a few suggestions. Based on the feedback, we added our ProviderLocations intersection table to our database outline to clearly show all tables that would be used in the final project. We also removed the totalAppointments attribute from the Patients and Providers tables to remove the risk of update anomalies for the static data. Additionally, this removes data redundancy as the Appointments table is also considered a transaction table, and totalAppointments can be tracked from the Appointments table with a suitable SQL query.

For the subsequent steps, we normalized our schema, created our data definition queries, added sample data to our database, designed our webpage UI, created data manipulation queries, and added stored procedures including a reset button. Again, the feedback received was predominantly positive with a majority of the suggestions on ways to improve our UI, such has using a more consistent CRUD operations layout between pages or better readability for page navigation titles.

One area that we consistently received feedback over many stages was regarding how we intended to delete data from our M:N relationship tables, Appointments and ProviderLocations. There were several suggestions to use On Delete Cascade to make deleting data a more simplified process. However, we remained steadfast in using On Delete Restrict functionality for our foreign keys. The reason being that we always envisioned our database to preserve historical appointment data and certain data, specifically appointment data, shouldn't be deleted without proper review by the database administrator. We have implemented several changes to make the web application easy to navigate and for the user to understand how to delete certain data with the restrict functionality in place.

Altogether, our web application demonstrates our ability to convert a business concept into a fully realized database through multiple planning, designing, and refining phases.

We made use of the Microsoft Copilot as a drafting assistant for stored procedures and error-messaging formatting in 'app.js'. Copilot was effective for generating boilerplate logic quickly but required further review as it sometimes made incorrect assumptions about our schema.

Overview

In the ever changing and expanding world of healthcare, one thing that remains constant is the need to efficiently document and manage patient information. This is especially true for OBGYN specialties in Oregon where according to Axios, the demand for abortion services alone increased by 14% between 2023 and 2024 after the overturning of Roe v. Wade in 2022 [1]. However, OBGYN services are not only limited to abortion care. A wide range of services fall under the OBGYN specialty umbrella including, as the specialty name alludes, OB or obstetric care. While Oregon Health Authority reports a decrease in general fertility rates from 60.0 births per 1,000 women ages 15-44 in 2010 to 45.6 births in 2024, inadequate care (defined as less than 5 prenatal visits during pregnancy or care that did not start until the third trimester) has increased from 5.3% to 6.4% in the same amount of time [2]. As there are several hospitals in rural Oregon that don't have labor and delivery units [3], and facilities are constantly navigating between network status with insurance companies, it is important that patients can find places to receive quality care.

Janeway's OBGYN Association has three clinics located in Oregon. Across the association, there are 12 physicians, 5 nurse practitioners, and 8 midwives who combined see over 1,800 patients per week. A website and database are needed for the Association in order to track the number of appointments patients are making and the number of appointments providers complete. This information can then be used to decide if or when another clinic location needs to be opened, or if additional providers need to be hired.

As previously stated, an online database is needed to track and preserve historical data, specifically relating to the number of appointments that have been completed. Because of the importance of the data, users cannot delete data without reviewing and possibly updating the data first. Additionally, as Janeway's OBGYN Association is a small organization servicing more rural parts of Oregon, when new clinics are added, they will be in a new city where none of the current clinics exist.

Database Outline

Patients: Records the details of the patients we serve.

- patientID: INT, auto_increment, unique, not NULL, PK
- firstName: VARCHAR(50), not NULL
- lastName: VARCHAR(50), not NULL
- dateOfBirth: DATE, not NULL
- address: VARCHAR(255), not NULL
- language: VARCHAR(50), not NULL
- insurancePayor: VARCHAR(50), not NULL
- Relationship: a 1:M relationship between Patients and Appointments is implemented with patientID as a FK inside of Appointments. This shows how patients can make multiple appointments.

AppointmentTypes: Tracks the different kinds of appointments a patient can schedule (e.g. New Patient, Procedure, etc.)

- typeID: VARCHAR(10), unique, not NULL, PK
- description: VARCHAR(255), not NULL
- durationInMinutes: INT, not NULL
- Relationship: a 1:M relationship between AppointmentTypes and Appointments is implemented with typeID as a FK inside of Appointments. This shows how there can be many appointments, but each appointment will only be one type.

Appointments: Records the details of appointments scheduled for patients.

- appointmentID: INT, auto_increment, unique, not NULL, PK
- apptDateTime: DATETIME, not NULL
- apptStatus: VARCHAR(50), not NULL,
- typeID: foreign key from AppointmentTypes entity
- patientID: foreign key from Patients entity
- providerID: foreign key from Providers entity
- clinicID: foreign key from Clinics entity
- Relationship: a transaction/fact table that connects both M:N relationship between Patients and Clinics as well as Patients and Providers. This is implemented by the patientID, providerID, and clinicID as FKs inside of Appointments. It shows how many patients can be seen by many providers, and many providers can see many patients. It also shows how patients can go to multiple clinic locations, and clinics can have many patients.

Providers: Records the details of the employed providers.

- providerID: INT, unique, not NULL PK
- firstName: VARCHAR(50), not NULL
- lastName: VARCHAR(50), not NULL
- startDate: DATE, not NULL
- title: VARCHAR(50), not NULL
- Relationship: a 1:M relationship between Providers and Appointments is implemented with providerID as a FK inside of Appointments. This shows how one provider can have many appointments that they complete.
- Relationship: a M:N relationship between Providers and Clinics is implemented using an intersection table titled, ProviderLocations with providerID as a FK. This shows how a provider can work at more than one clinic location.

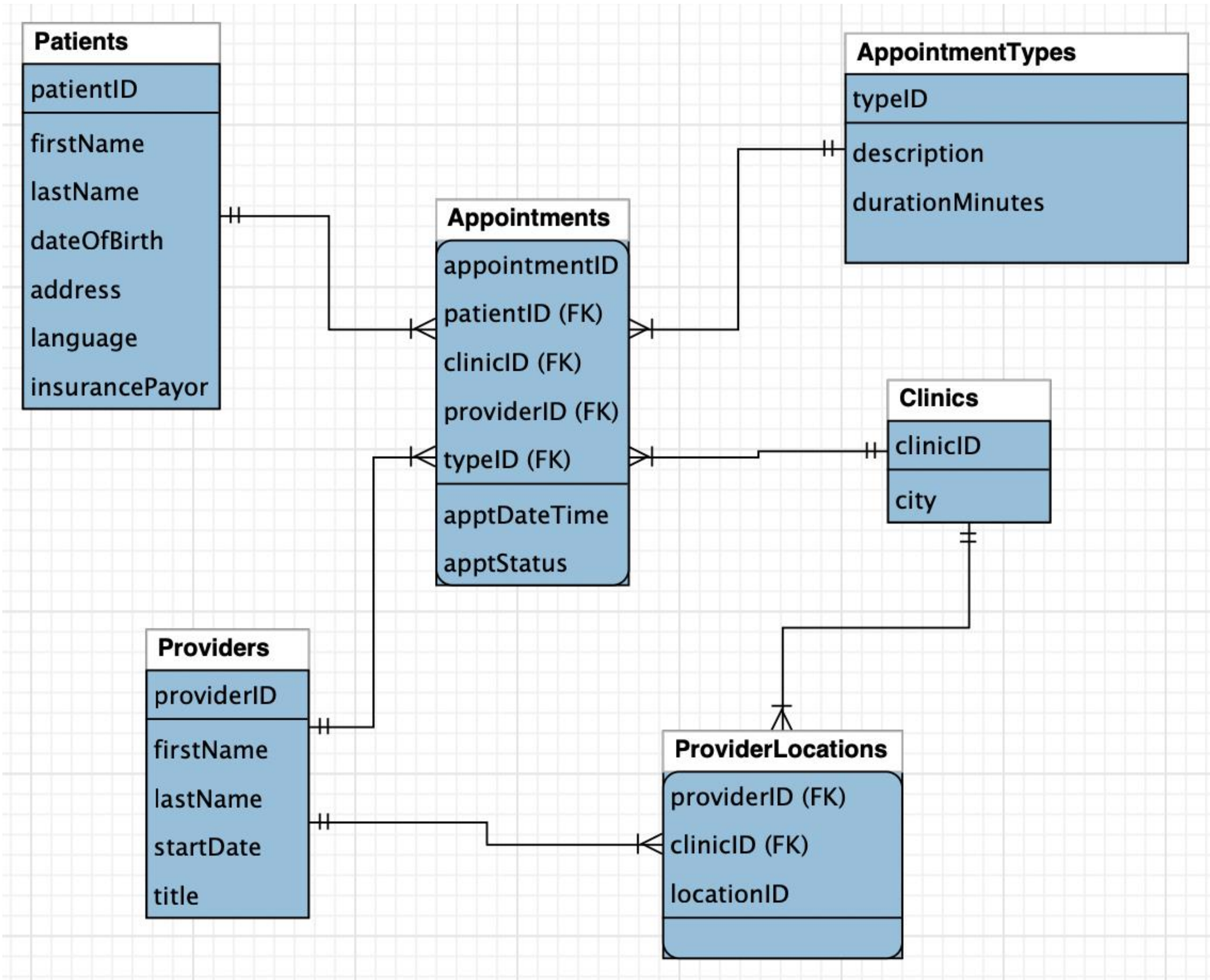
Clinics: Lists the clinic location information.

- clinicID: INT, auto_increment, unique, not NULL, PK
- city: VARCHAR(50), not NULL
- Relationship: a 1:M relationship between Clinics and Appointments is implemented with clinicID as a FK inside of Appointments.
- Relationship: a M:N relationship between Clinics and Providers is implemented using an intersection table title, ProviderLocations, with clinicID as a FK. This shows how a clinic can have many providers working at that location.

ProviderLocations: An intersection table.

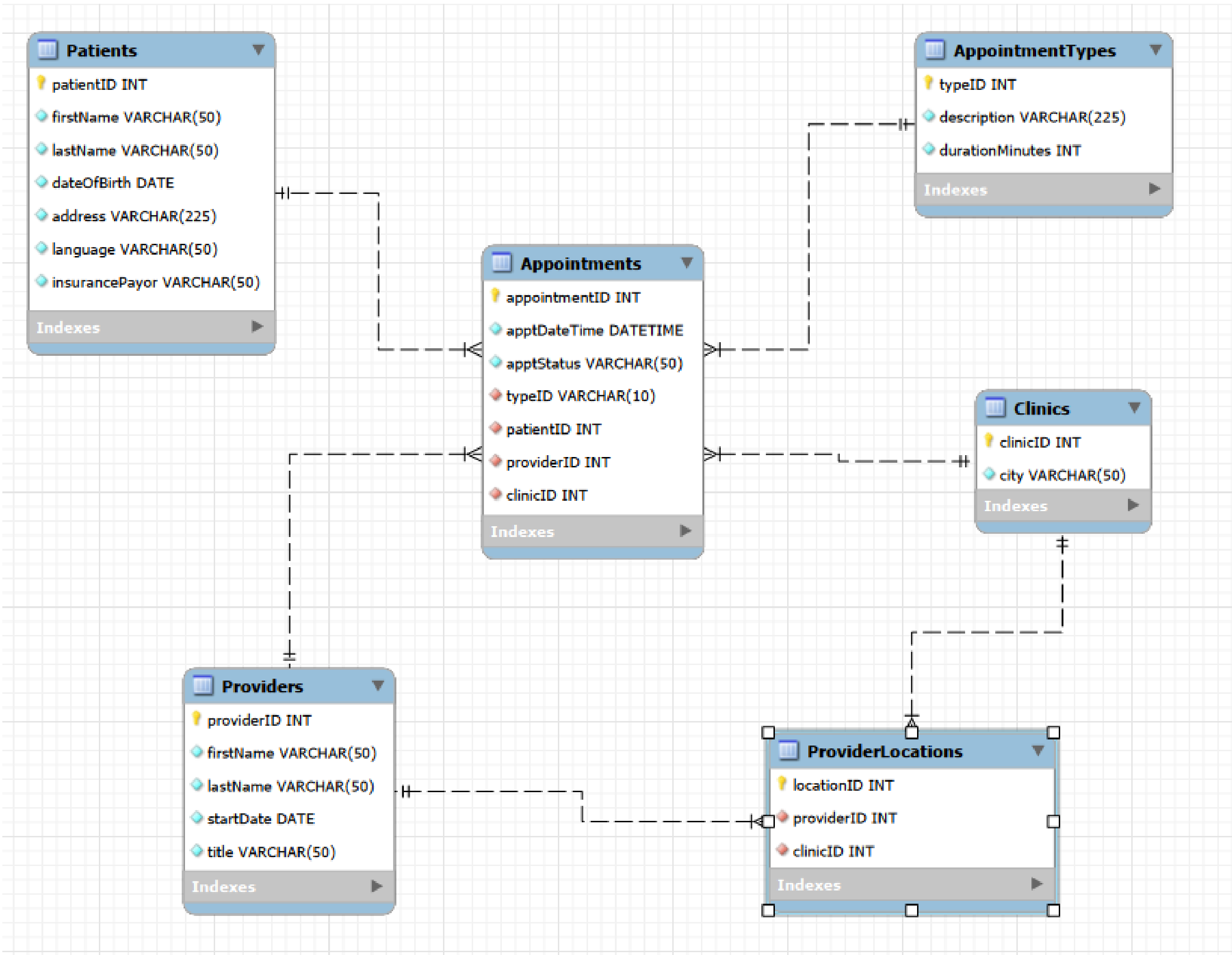
- locationID: INT, auto_increment, unique, not NULL, PK
- providerID: foreign key from Providers entity.
- clinicID: foreign key from Clinics entity.
- Relationship: This intersection table is automatically created between the M:N relationship between Clinics and Providers showing how a provider can work at multiple clinic locations, and a clinic location can have multiple providers.

ER Diagram



Patients
patientID
firstName
lastName
dateOfBirth
address
language
insurancePayor
AppointmentTypes
typeID
description
durationMinutes
Appointments
appointmentID
patientID (FK)
clinicID (FK)
providerID (FK)
typeID (FK)
apptDateTime
apptStatus
Clinics
clinicID
city
Providers
providerID
firstName
lastName
startDate
title
ProviderLocations
providerID (FK)
clinicID (FK)
locationID

Schema



Patients
patientID INT
firstName VARCHAR(50)
lastName VARCHAR(50)
dateOfBirth DATE
address VARCHAR(225)
language VARCHAR(50)
insurancePayor VARCHAR(50)
Indexes
AppointmentTypes
typeID INT
description VARCHAR(225)
durationMinutes INT
Indexes
Appointments
appointmentID INT
apptDateTime DATETIME
apptStatus VARCHAR(50)
typeID VARCHAR(10)
patientID INT
providerID INT
clinicID INT
Indexes
Clinics
clinicID INT
city VARCHAR(50)
Indexes
Providers
providerID INT
firstName VARCHAR(50)
lastName VARCHAR(50)
startDate DATE
title VARCHAR(50)
Indexes
ProviderLocations
locationID INT
providerID INT
clinicID INT
Indexes

Example Data

Patients – Examples of patients served and their information.

patientID	firstName	lastName	dateOfBirth	address	language	insurancePayor
1	Nyota	Uhura	1972-03-26	1701 Enterprise Drive Corvallis, OR 97330	Swahili	United Healthcare
2	Jadzia	Dax	1988-01-01	10 Forward Avenue Corvallis, OR 97330	English	Self-pay
3	B'Elanna	Torres	1986-06-06	P.O. Box 709 Salem, OR 97301	Spanish	Blue Cross Blue Shield
4	Kira	Nerys	1991-09-15	74205 Promenade Lane Salem, OR 97301	English	Blue Cross Blue Shield
5	Beckett	Mariner	2000-04-19	24593 Federation Drive Eugene, OR 97401	English	Aetna

AppointmentTypes – Examples of the different types of appointments patients can schedule.

typeID	description	durationInMinutes
NOB	New OB	30
NP	New Patient	30
OBR	OB Regular	15
OVR	Office Visit Regular	15
PROC	Procedure	45

Appointments – Examples of appointments.

appointmentID	apptDateTime	apptStatus	typeID	patientID	providerID	clinicID
1	2026-07-15 14:30:00	Scheduled	NP	2	113	1
2	2026-03-31 13:00:00	Completed	OVR	4	102	2
3	2026-04-28 09:30:00	Cancelled	OVR	1	249	1
4	2025-12-10 10:00:00	Completed	NOB	5	368	3
5	2026-08-09 15:00:00	Scheduled	PROC	3	113	2

Providers – Examples of employed providers within the clinics.

providerID	firstName	lastName	startDate	title
102	Katherine	Pulaski	2015-11-11	Doctor
113	Beverly	Crusher	2020-10-16	Doctor
249	Christine	Chapel	2017-03-06	Nurse Practitioner
368	Julienne	Bashir	2021-02-12	Midwife

Clinics – Examples of the clinic locations, currently the association has three locations.

clinicID	city
1	Corvallis
2	Salem
3	Eugene

ProviderLocations – Example of intersection table that tracks locations providers work at.

providerID	clinicID
102	2
113	1
113	2
249	1
368	3

UI Screen Captures

Figure 1: Read/Update/Delete data on Patients page

Browse Existing Patients

Displays list of existing patients and their information.

Patient ID	First Name	Last Name	Date of Birth	Address	Language	Insurance Payor	
1	Nyota	Uhura	1972-03-26	1701 Enterprise Drive Corvallis, OR 97330	Swahili	United Healthcare	Edit Delete
2	Jadzia	Dax	1988-01-01	10 Forward Avenue Corvallis, OR 97330	English	Self-pay	Edit Delete
3	B'Elanna	Torres	1986-06-06	P.O. Box 709 Salem, OR 97301	Spanish	Blue Cross Blue Shield	Edit Delete
4	Kira	Nerys	1991-09-15	74205 Promenade Lane Salem, OR 97301	English	Blue Cross Blue Shield	Edit Delete
5	Beckett	Mariner	2000-04-19	24593 Federation Drive Eugene, OR 97401	English	Aetna	Edit Delete

Figure 2: Create New on Patients page

Add A New Patient

Add a new patient to the database by filling out the form below.

First Name:

Last Name:

Date of Birth:

mm/dd/yyyy

Address:

Language:

Insurance Payor:

Add Patient

Figure 3: Read/Update/Delete data on Appointment Types page

Browse Appointment Types

Displays the appointment type catalog used when scheduling appointments.

Type ID	Description	Duration (minutes)	
NOB	New OB	30	Edit Delete
NP	New Patient	30	Edit Delete
OBR	OB Regular	15	Edit Delete
OVR	Office Visit Regular	15	Edit Delete
PROC	Procedure	45	Edit Delete

Figure 4: Create New on Appointment Types page

Create an Appointment Type

Type ID:

Description:

Duration (minutes):

Create Appointment Type

Figure 5: Read/Update/Delete data on Providers page

Browse Providers

Displays the list of providers and their information.

Provider ID	First Name	Last Name	Start Date	Title	
102	Katherine	Pulaski	2015-11-11	Doctor	Edit Delete
113	Beverly	Crusher	2020-10-16	Doctor	Edit Delete
249	Christine	Chapel	2017-03-06	Nurse Practitioner	Edit Delete
368	Julienne	Bashir	2021-02-12	Midwife	Edit Delete

Figure 6: Create New on Providers page

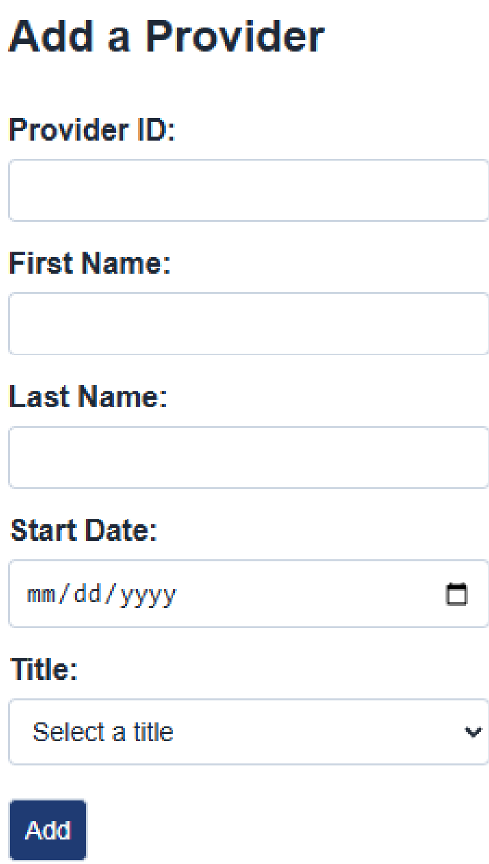


Figure 7: Read/Update/Delete data on Clinics page

Browse Clinics

Displays all clinic locations in the system.

Clinic ID	City	
1	Corvallis	Edit Delete
2	Salem	Edit Delete
3	Eugene	Edit Delete

Figure 8: Create New on Clinics page

Add a Clinic

City:

Create Clinic

Figure 9: Read/Update/Delete data on Appointments page (M:N relationship with Patients, Appointment Types, Providers and Clinics)

Browse Appointments

Displays all appointment records in the system.

AppointmentID	Appointment Date/Time	Appointment Status	Type ID	Patient ID	Provider ID	Clinic ID	
1	2026-07-15 14:30:00	Scheduled	NP	2	113	1	Edit Delete
2	2026-03-31 13:00:00	Completed	OVR	4	102	2	Edit Delete
3	2026-04-28 09:30:00	Cancelled	OVR	1	249	1	Edit Delete
4	2025-12-10 10:00:00	Completed	NOB	5	368	3	Edit Delete
5	2026-08-09 15:00:00	Scheduled	PROC	3	113	2	Edit Delete

Figure 10: Create New on Appointments page (M:N relationship with Patients, Appointment Types, Providers and Clinics)

Create a New Appointment

Appointment Date/Time:

mm/dd/yyyy --:-- --

Appointment Status:

Scheduled

Appointment Type:

Select an appointment type

Patient:

Select a patient

Provider:

Select a provider

Clinic:

Select a clinic

Create Appointment

Figure 11: Read/Update/Delete data on Provider Locations page (M:N relationship between Clinics and Providers)

Browse Provider Locations

Displays which providers work at which clinic locations.

Location ID	Provider ID	Clinic ID	First Name	Last Name	City	
1	113	1	Beverly	Crusher	Corvallis	Edit Delete
2	113	2	Beverly	Crusher	Salem	Edit Delete
3	249	1	Christine	Chapel	Corvallis	Edit Delete
4	368	3	Julienne	Bashir	Eugene	Edit Delete
5	102	2	Katherine	Pulaski	Salem	Edit Delete

Figure 12: Create New on Provider Locations page (M:N relationship between Clinics and Providers)

Add a Provider Location

Provider:

Select a provider

Clinic:

Select a clinic

Add Provider Location

Citations

[1] M. Gebel, "Demand for OBGYN services rises in Oregon" *Axios Portland*, March 13, 2024 [Online]. Available: https://www.axios.com/local/portland/2024/03/13/demand-obgyn-abortion-rises-oregon-idaho Accessed April 07, 2026

[2] "Oregon Annual Trends in Birth & Pregnancy, 2010-2024" *Oregon Health Authority,* Dec. 24, 2025 [Online]. Available: https://visual-data.dhsoha.state.or.us/t/OHA/views/Annualtrendsinbirthandpregnancydashboard/TrendsDashboard?%3AisGuestRedirectFromVizportal=y&%3Aembed=y Accessed April 07, 2026

[3] M. Fass, "Gov. Tina Kotek announces funding for maternity care in rural hospitals, as some labor and delivery units close their doors" *OPB,* Feb. 09, 2026 [Online]. Available: https://www.opb.org/article/2026/02/09/think-out-loud-maternity-care-rural-hospitals/#:~:text=Ten%20of%20Oregon's%2034%20rural,risk%20in%20cases%20of%20emergency. Accessed April 07, 2026

[4] M. Curry, Exploration - Database Design Patterns, [Online]. Available: https://canvas.oregonstate.edu/courses/2042369/pages/exploration-database-design-patterns#slides Accessed April 14, 2026